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Music Genre Classification

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ABSTRACT

Music Genre Classification Model is a model to classify songs or an audio music based on variety of features into the corresponding genre. This project aims at developing a machine learning and deep learning models and train them on a dataset of audio features extracted from a large collection of music tracks. The models are designed and developed to produce a better accuracy for Music Genre Classification.

EXISTING SYSTEM

The Existing system includes different models to automatically classify different musical genres from dataset. It uses Label Encoder to convert class label of categorical type to numerical type. It compares the performances of Random Forest and XG Boost models and logged their results in terms of prediction accuracies. XG Boost has given the highest accuracy amongst all other models.

Literature Survey

S. No	Authors	Description
1	Prajwal, R., et al. "Music genre classification using machine learning." Int. Res. J. Mod. Eng.	In this paper, four machine learning models knn, SVM, Logistic Regression, and random forest are used. They employ different techniques to find the best combination of hyperparameters on which models would perform optimally which improves accuracy. So, by using the techniques & finding the optimal K value for k-nearest neighbors.

S. No	Authors	Description
2	Fan, Minglei. "Application of music industry based on the deep neural network." Scientific Programming 2022 (2022): 1-6	In this study, we use a novel deep convolutional neural network (CNN) to extract multiple features from music signals and classify them. The original spectrogram is used as the input of the CNN. Finally, the network structure of the CNN is designed, and the effect of different parameters on the recognition rate is investigated.

S. No	Authors	Description
3	Shah, Mitt, et al. "Music genre classification using deep learning." 2022 6th(ICCMC). IEEE, 2022.	To predict the genre of the audio signal, models such as Support Vector Machines (SVM), Random Forests, XGBoost and CNN are used. The GTZAN dataset was used for model training and testing. Machine learning and deep learning models each had their own set of features. A comparison analysis is proposed between these models, demonstrating that CNN outperforms machine learning models.

PROPOSED SYSTEM

- The proposed system generates Mel-spectrogram images from different audio tracks.
- These generated images are trained using Deep Learning models for better prediction.
- This includes deep learning models like transfer and CNN.

TOOLS AND TECHNOLOGIES

Libraries:

pandas, matplotlib, seaborn, numpy, Scikit-learn, Tensor flow, librosa

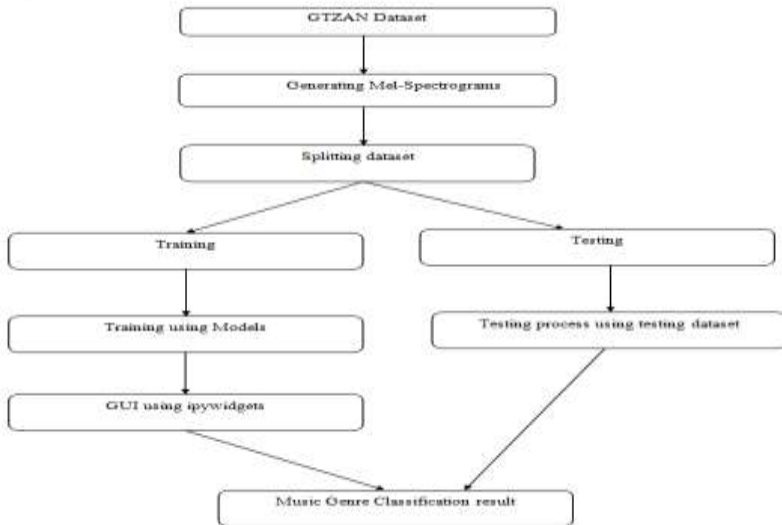
Models:

Decision Tree Classifier, Multinomial Logistic Regression, RandomForest Classifier, XGBoost Classifier, Transfer, CNN

GUI Development:

ipywidgets

WORKFLOW



Methodology

Decision Tree Classifier:

It is a tree-like structure where each internal node denotes the feature, branches denote the rules and the leaf nodes denote the result of the algorithm.

Random Forest Classifier:

It is an ensemble method, meaning that a random forest model is made up of a large number of small decision trees, called estimators, which each produce their own predictions.

Multinomial Logistic Regression:

It is a classification method that generalizes logistic regression to multiclass problems. It is a model that is used to predict the probabilities of the different possible outcomes of a categorically distributed dependent variable.

XG Boost:

XGBoost is a scalable and highly accurate implementation of gradient boosting that pushes the limits of computing power for boosted tree algorithms, being built largely for energizing machine learning model performance and computational speed.



Transfer Model:

Transfer learning is a technique where a pre-trained model on one task is used as a starting point for a model on a different but related task i.e, instead of training a model from scratch, you take a pre-trained model (already good at recognizing certain patterns) and fine-tune it for your specific task.

Convolutional Neural Network:

Is deep learning neural network designed for processing structured arrays of data such as images. CNN are widely used in computer vision and have become the state of the art for many visual applications such as image classification. It helps in understanding features that helps in developing more accurate Music Genre Classification model.

Comparitive Analysis

Models implemented using dataset features:

S.No	Models	Accuracy
1	Decision Tree	48.44%
2	Multinomial Logistic Regression	54.90%
3	Random Forest	69.21%
4	XG Boost	72.67%

Models implemented using audio tracks:

S. No	Models	Accuracy
1	Transfer	62.99%
2	CNN	64.99%



Output

Audio Path: /content/drive/MyDrive/mlproject/genres_original/classical/classical.00002.wav

Predict

```
1/1 [=====] - ETA: 0s  
1/1 [=====] - 0s 24ms/step  
Model Prediction: classical
```

DATASET

GTZAN is a public dataset collected from different sources. It's a collection of 10 genres and each genre consists of 100 audio files and its corresponding image files and also consists of 1000 rows and 60 columns where each column represent unique feature and is approximately 1.2GB size.

REFERENCES

- [1] Prajwal, R., et al. "Music genre classification using machine learning." Int. Res. J. Mod. Eng. Technol. Sci 3.7 (2021): 953-957.
- [2] Fan, Minglei. "Application of music industry based on the deep neural network." Scientific Programming 2022 (2022): 1-6.
- [3] Shah, Mitt, et al. "Music genre classification using deep learning." 2022 6th International Conference on Computing Methodologies and Communication (ICCMC). IEEE, 2022.

THANK YOU